T1 · Probability & concentration

Q1 · Error reduction for decision problems

Theme of MIT 6.889 HW0 · difficulty ★

A randomized algorithm A for a decision problem errs with probability at most $1/4$ on every input and runs in time T. Design B that errs with probability at most β and runs in time $O(T \log(1/\beta))$. Prove it.

Solution. Run A independently k times and output the majority answer. Let X be the number of wrong runs; X is a sum of independent indicators with $\mu = \mathbb{E}X \le k/4$. B errs only if $X \ge k/2$. Using the upper-tail Chernoff bound with $\mu_H = k/4$ and $\delta = 1$: $\Pr[X \ge k/2] \le e^{-\mu_H/3} = e^{-k/12}$. Choosing $k = \lceil 12 \ln(1/\beta) \rceil$ gives error $\le \beta$ and time $O(kT) = O(T \log(1/\beta))$.

Q2 · Coupon collector

Theme of MIT 6.889 HW0 and L1 balls-into-bins · ★

A fair n-sided die is rolled repeatedly. (a) Show that the expected number of rolls until every face has appeared is $nH_n = n \ln n + O(n)$. (b) Show that after $2n \ln n$ rolls, every face has appeared with probability at least $1 - 1/n$.

Solution. (a) Split the process into phases: phase i ($i = 0, \dots, n-1$) lasts while exactly i distinct faces have been seen. Each roll ends phase i with probability $\frac{n-i}{n}$, so its length is geometric with mean $\frac{n}{n-i}$. By linearity, $\mathbb{E}[\text{total}] = \sum_{i=0}^{n-1} \frac{n}{n-i} = n \sum_{j=1}^{n} \frac{1}{j} = nH_n$. (b) A fixed face is missed in all $2n \ln n$ rolls with probability $(1 - \frac{1}{n})^{2n \ln n} \le e^{-2 \ln n} = n^{-2}$. Union bound over n faces: some face is missing with probability $\le n \cdot n^{-2} = 1/n$.

Q3 · Average number of hashtags

Theme of NUS CS5234 (2019) Tutorial 4 · ★★

A stream of tweets arrives; each tweet contains between 0 and 70 hashtags. Using a uniform sample of s tweets drawn *with replacement*, estimate the average number A of hashtags per tweet within additive error ε, with probability at least $2/3$. How large must s be? What changes if you want error εA instead and know $A \ge 1$?

Solution. Let $Y_j \in [0, 70]$ be the hashtag count of the j-th sample and $Z_j = Y_j/70 \in [0, 1]$, independent with mean $A/70$. Additive error ε on $\bar{Y}$ is additive error $\varepsilon/70$ on $\bar{Z}$. Hoeffding: $\Pr[|\bar{Z} - \frac{A}{70}| \ge \frac{\varepsilon}{70}] \le 2e^{-2s\varepsilon^2/70^2} \le \frac{1}{3}$ for $s = \frac{70^2 \ln 6}{2\varepsilon^2} = O(1/\varepsilon^2)$ (the constant is about 4400). *Relative error*: $\sum Z_j$ has $\mu = sA/70 \ge s/70$; Chernoff with $\delta = \varepsilon$ gives $2e^{-\varepsilon^2 \mu/3} \le 2e^{-\varepsilon^2 s/210} \le \frac{1}{3}$ for $s = \frac{210 \ln 6}{\varepsilon^2} \approx \frac{377}{\varepsilon^2}$. Knowing a lower bound on A is what makes relative error cheap. *Streaming*: the sample itself can be maintained with reservoir sampling (s independent single-item reservoirs give a sample with replacement).

T2 · Query algorithms & lower bounds

Q4 · Boosting an approximation scheme

Theme of MIT 6.889 HW0 · ★

Algorithm A outputs a $(1+\varepsilon)$-approximation of a positive quantity V with probability $3/4$. Build B that outputs a $(1+\varepsilon)$-approximation with probability $1 - \delta$, using $O(\log(1/\delta))$ calls to A. Why does taking the average of the runs not work?

Solution. Run A k times and output the median. Call a run good if its output lies in $[V/(1+\varepsilon), (1+\varepsilon)V]$. If the median is bad, then at least $k/2$ runs are bad. The number of bad runs has mean $\leq k/4$; by Chernoff (upper tail, $\mu_H = k/4$, $\delta = 1$) it reaches $k/2$ with probability $\leq e^{-k/12}$. Take $k = \lceil 12 \ln(1/\delta) \rceil$. *Why not the average*: a bad run can be arbitrarily wrong (e.g. output $10^9 V$), so one bad run can ruin the average; the median only needs a majority of good runs.

Q5 · Element distinctness needs $\Omega(n)$

Reduction pattern of L3–L4 · ★

Given query access to $a_1, \ldots, a_n$ (n even), decide exactly whether all values are distinct. Prove that the randomized query complexity is $\Omega(n)$ by a reduction from OR.

Solution. Given an OR instance $b \in \{0,1\}^{n/2}$, define $a_{2j-1} = 2j-1$ and $a_{2j} = 2j$ if $b_j = 0$, $a_{2j} = 2j-1$ if $b_j = 1$. *Correctness*: the only possible repeated value is inside pair j, which happens iff $b_j = 1$; so the array is distinct iff $\mathrm{OR}(b) = 0$. *Simulation*: querying a_{2j-1} needs no OR query; querying a_{2j} needs one query to b_j ($C = 1$). Hence $R(\text{Distinct}) \geq R(\mathrm{OR}_{n/2}) \geq \frac{n/2}{3} = \Omega(n)$.

Q6 · Finding the maximum needs $\Omega(n)$

Yao's principle for a search problem · ★★

Given query access to $a_1, \ldots, a_n$, output an index of a maximum entry, correct with probability $\geq 2/3$. Prove that $\Omega(n)$ queries are needed. *Hint: the answer is not binary, but Yao still applies.*

Solution. Let μ: all entries are 0 except a 1 at a uniformly random position P; the unique correct answer is P. Fix a deterministic algorithm with q queries. Along the all-zero path it queries fixed positions $p_1, \ldots, p_q$. If P is among them (probability q/n) it may answer correctly. Otherwise it sees only zeros, its output is one fixed index, which equals P with probability at most $\frac{1}{n-q}$ given that P is among the $n-q$ unqueried positions. Success $\leq \frac{q}{n} + \frac{n-q}{n} \cdot \frac{1}{n-q} = \frac{q+1}{n}$. For this to be $\geq \frac{2}{3}$ we need $q \geq \frac{2n}{3} - 1$. Yao's argument does not use that the output is binary, so $R \geq D_\mu = \Omega(n)$.

Q7 · Average degree with additive error

Theme of L2 · ★

In the adjacency-list model (degree queries), estimate the average degree $\bar{d}$ of an n-vertex graph within additive error εn with probability $2/3$. How many queries suffice? Why does the same idea fail for a $(1 \pm \varepsilon)$-approximation of $\bar{d}$ on sparse graphs?

Solution. Sample s vertices with replacement and average $d_v/n \in [0,1]$. Hoeffding: $s = O(1/\varepsilon^2)$ gives additive error ε on the scaled average, i.e. εn on $\bar{d}$. For a $(1 \pm \varepsilon)$-approximation, the relative variance can be huge: in a star, $\bar{d} \approx 2$, but a random vertex has degree $n-1$ only with probability $1/n$, so $\mathrm{Var}(d_v)/\bar{d}^2 = \Theta(n)$ and Chebyshev needs $\Theta(n/\varepsilon^2)$ samples; indeed with $o(n)$ samples one usually never sees the centre. L2 fixes this by charging each edge to its lower-ranked endpoint, giving $O(\sqrt{n}/\varepsilon^2)$.

T3 · Property testing

Q8 · Testing palindromes

Generic "far ⇒ many witnesses" pattern · ★

A string $x \in \{0,1\}^n$ is a palindrome if $x_i = x_{n+1-i}$ for all i. (a) What is the distance of x from being a palindrome? (b) Give a one-sided tester with $O(1/\varepsilon)$ queries.

Solution. (a) Pair position i with $n+1-i$; there are $\lfloor n/2 \rfloor$ pairs. Each mismatched pair needs exactly one change and matched pairs need none, so the distance equals the number M of mismatched pairs. (b) If x is ε-far then $M \geq \varepsilon n$, so a uniformly random pair is mismatched with probability $\geq \frac{\varepsilon n}{n/2} = 2\varepsilon$. Sample $k = \lceil 1/\varepsilon \rceil$ random pairs (2 queries each) and reject iff some pair mismatches. Palindromes are always accepted; an ε-far string is accepted with probability $\leq (1-2\varepsilon)^k \leq e^{-2} < \frac{1}{3}$.

Q9 · Testing connectivity in bounded-degree graphs

Theme of MIT 6.889 HW1 (Goldreich–Ron) · ★★★

A graph has maximum degree d and is given by adjacency lists. It is ε-far from connected if more than εdn edge modifications are needed to make it connected. (a) Show that an ε-far graph has more than εdn connected components (you may assume that adding one edge can always merge two components while keeping the degree bound, possibly after removing one edge). (b) Deduce that more than $\frac{\varepsilon dn}{2}$ components have at most $\frac{2}{\varepsilon d}$ vertices. (c) Give a one-sided tester whose query complexity is independent of n.

Solution. (a) With c components, $c-1$ merges (each $O(1)$ modifications; with the stated assumption, one) make the graph connected, so ε-far forces $c - 1 > \varepsilon dn$ up to the constant. (b) Fewer than $\frac{\varepsilon dn}{2}$ components can have more than $\frac{2}{\varepsilon d}$ vertices, since together they would contain more than n vertices. So more than $\varepsilon dn - \frac{\varepsilon dn}{2} = \frac{\varepsilon dn}{2}$ components are small. (c) The small components contain more than $\frac{\varepsilon dn}{2}$ vertices in total, so a random vertex lies in a small component with probability $> \frac{\varepsilon d}{2}$. Sample $s = \lceil \frac{4}{\varepsilon d} \rceil$ vertices; from each run BFS until it has found $\frac{2}{\varepsilon d} + 1$ vertices or exhausted its component. Reject iff some BFS exhausts a component with fewer than n vertices. A connected graph is never rejected. For an ε-far graph, the probability no sample hits a small component is $\leq (1 - \frac{\varepsilon d}{2})^s \leq e^{-2}$. Each BFS explores $O(\frac{1}{\varepsilon d})$ vertices with $\leq d$ queries each, so the total is $O\big(\frac{1}{\varepsilon^2 d}\big)$ queries, independent of n.

Q10 · Distance between distributions

Theme of L4 and MIT 6.889 HW3 · ★★

(a) Show $\Delta_{\mathsf{TV}}(\mu, \nu) = \max_{A \subseteq [n]} |\mu(A) - \nu(A)|$. (b) Let μ be uniform on $[n]$ and ν uniform on a fixed set S of size $n/4$. Compute $\Delta_{\mathsf{TV}}(\mu, \nu)$ and $\|\nu\|_2^2$. (c) How many samples does the collision tester of L4 need to tell them apart, up to constants?

Solution. (a) Let $A^* = \{i : \mu(i) > \nu(i)\}$. For any A, $\mu(A) - \nu(A) \leq \sum_{i \in A \cap A^*} (\mu(i) - \nu(i)) \leq \mu(A^*) - \nu(A^*)$, and similarly for $\nu(A) - \mu(A)$ with the complement. Since both distributions sum to 1, $\mu(A^*) - \nu(A^*) = \nu(\bar{A^*}) - \mu(\bar{A^*}) = \frac{1}{2} \sum_i |\mu(i) - \nu(i)|$. (b) Take $A = S$: $\nu(S) - \mu(S) = 1 - \frac{1}{4} = \frac{3}{4}$, so $\Delta_{\mathsf{TV}} = \frac{3}{4}$. $\|\nu\|_2^2 = \frac{n}{4} \cdot (\frac{4}{n})^2 = \frac{4}{n}$, four times the uniform value $\frac{1}{n}$. (c) Here the collision probabilities differ by a constant factor, so a constant-accuracy estimate of $\|\cdot\|_2^2$ suffices: $\Theta(\sqrt{n})$ samples (about $\sqrt{n}$ are needed just to see any collision).

T4 · Streaming

Q11 · Morris++

Theme of NUS CS5234 (2019) Tutorial 4 · ★★

Using the Morris counter (unbiased, Var $< A^2$), design a streaming algorithm that outputs $\hat{A} \in [(1-\varepsilon)A, (1+\varepsilon)A]$ with probability $1 - \delta$. What is its space?

Solution. *Morris+*: average $t = \lceil 3/\varepsilon^2 \rceil$ independent counters; Var $< A^2/t$, so Chebyshev gives $\Pr[|\bar{A} - A| > \varepsilon A] < \frac{1}{t\varepsilon^2} \leq \frac{1}{3}$. *Morris++*: run $m = O(\log(1/\delta))$ independent copies of Morris+ and output the median; by the median trick the failure probability is $\leq \delta$. Each counter stores $C \approx \log A$, i.e. $O(\log \log n)$ bits (a counter that exceeds $O(\log n)$ can be capped; this happens with tiny probability). Total space $O\big(\frac{1}{\varepsilon^2} \log \frac{1}{\delta} \log \log n\big)$ bits.

Q12 · Weighted reservoir sampling

Classic variant of L5 reservoir sampling · ★★

Items $x_1, x_2, \ldots$ arrive with positive weights $w_1, w_2, \ldots$. Keep one item so that at the end x_i is kept with probability w_i/W_n, where $W_i = \sum_{j \le i} w_j$. Give the algorithm and prove it.

Solution. Maintain W and the current sample. When x_i arrives, update $W \leftarrow W_i$ and replace the sample by x_i with probability w_i/W_i. *Proof*: x_i is chosen at step i with probability $\frac{w_i}{W_i}$ and survives step $j > i$ with probability $1 - \frac{w_j}{W_j} = \frac{W_{j-1}}{W_j}$. The product telescopes: $\frac{w_i}{W_i} \cdot \frac{W_i}{W_{i+1}} \cdots \frac{W_{n-1}}{W_n} = \frac{w_i}{W_n}$. With all weights equal to 1 this is the L5 algorithm. Space: the sample plus the running total.

Q13 · Cycles and spanners in a stream

Theme of L5 · ★★

Edges of an n-vertex graph arrive in a stream. (a) Decide whether the graph contains a cycle using $O(n \log n)$ bits. (b) Run the greedy spanner with $k = 2$. Show that the output H has no cycle of length ≤ 4 and that $d_H(u, v) \le 3\, d_G(u, v)$. (c) Why is $|E(H)| = O(n^{3/2})$?

Solution. (a) Maintain a spanning forest with union–find. The graph has a cycle iff some arriving edge joins two vertices already connected in the forest; report that. The forest has $\le n - 1$ edges, so $O(n \log n)$ bits. (b) An edge is added only if it closes no cycle of length ≤ 4 in H, and later additions cannot create such a cycle through an earlier edge without that later edge closing it; so H has girth ≥ 5. A rejected edge (u, v) closed a cycle of length ≤ 4, so H already had a u–v path of length ≤ 3. Replacing each edge of a shortest u–v path in G by such a path gives $d_H \le 3 d_G$. (c) By the L5 lemma with $k = 2$: girth > 4 implies that the first two BFS layers from any vertex form a tree, so $(d_{\min} - 1)^2 \le n$; peeling low-degree vertices extends this to the average degree, giving $|E(H)| = O(n^{1+1/2}) = O(n^{3/2})$.